Interactive Learning Assistant for Children

Find the Object | Eye Tracking vs. Hand Gestures

An HCI application-design proposal for children aged 5–10, comparing two camera-based selection techniques within the same learning activity.

Team members and section responsibilities

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The report distinguishes prototype screen representations and planned functionality from implemented systems or measured research results.

Evidence and scope

At Phase I, the team has prepared scenarios, provisional personas, a proposed survey, Figma interface screens and a conceptual interaction architecture. The real-world object detector, gaze calibration, hand tracking, sensor connections and performance measurements have not yet been implemented or empirically validated.

0. Project overview and shared specification

The Interactive Learning Assistant for Children is a proposed educational system for ages 5–10. In the "Find the Object" activity, an object-facing camera captures familiar physical items and the interface shows a video view with tracked object regions. The child sees a short prompt such as "Can you find the apple?" and selects the corresponding region using one of two alternative techniques.

Dimension	Eye Tracking	Hand Gestures
Child action	Look at the on-screen target object for approximately 1.5 seconds (provisional dwell).	Move a hand-controlled pointer over the on-screen target and perform a pinch gesture.
Required sensing	Child-facing gaze estimation and calibration; mapping gaze to a tracked displayed object.	Child-facing hand landmarks and gesture recognition; pointer-to-target matching.
Shared behavior	Same object feed, prompts, selection validation, feedback, retries and five-question structure.	Same object feed, prompts, selection validation, feedback, retries and five-question structure.

0.1 Intended interaction boundaries

- A proposed child-facing camera supports gaze/hand sensing; an object-facing camera captures the real objects. Two-camera operation remains a feasibility assumption.
- Gaze selects an object shown in the screen's camera view; it does not directly determine which physical object in three-dimensional room space the child is looking at.
- Adult assistance is permitted for initial permissions, setup and recovery. Mouse, keyboard and touchscreen are not intended primary gameplay inputs for the child.
- A five-question session with three visible example objects (apple, cup and book) is used for the scenario. Object availability and recognition must be tested later.
- A completed activity means five questions completed (5/5). First-attempt correctness is a separate potential metric. Illustrative progress values in Figma are not measured data.
- Spoken instructions and audio responses are intended alongside visual feedback, but are not operating features of the Figma prototype.

0.2 Future evaluation objective

The proposed study will compare the modes under matched activities and conditions, recording setup time, task time, first-attempt correctness, accidental selections, retries, tracking failures, observed engagement and child-appropriate satisfaction measures. No empirical results are presented in this Phase I report.

1. Scenario & User Journey

Section author: Mostafa Mohamed (249141)

1.1 Main child scenario: Nour

Nour is a seven-year-old girl in Grade 2. On a Saturday afternoon, she sits at the family desk in the living room in front of a laptop, with her mother beside her. Her mother wants Nour to spend fifteen minutes on a learning game, and Nour finds worksheets boring but enjoys games where she can move and look around. The session is meant to help her practise recognising and naming everyday objects.

Before the game starts, her mother completes a one-time setup. She places three familiar items, an apple, a cup, and a book, on a plain tablecloth where the object-facing camera can see them, and she allows camera access when the browser asks. These three objects are only an example; the final set of supported objects will depend on the detection model the team chooses and tests later. Nour sits about an arm's length from the screen so that her face and hands are visible to the child-facing camera. No account or sign-in is needed.

The Welcome screen shows the title "Let's Learn and Play!" and one large START LEARNING button. Because gaze calibration and the hand tutorial have not happened yet, her mother presses it for her. The next screen, "How Would You Like to Play?", shows two large cards: Hand Gestures ("Point and pinch to select") and Eye Tracking ("Look at an object to select it"). A line under the cards says that an adult can help. Nour likes the eye picture, so she picks the Eye Tracking card, and her mother confirms the choice. The "Let's Get Ready!" screen asks her to look at the dots to set up eye tracking. A target appears in five places one after another, the label counts "Step 1 of 5" and so on, and a progress bar fills. When the green message "Calibration Complete! Your eyes are ready to play." appears, Nour looks at CONTINUE.

The "Let's Get Things Ready!" screen shows a live preview of the table with a "Camera Ready" check and three short tips: place the objects where the camera can see them, keep them apart, and make sure they are well lit. When "Everything Looks Good!" appears, Nour looks at START GAME. The game screen shows the live view with a coloured box around the apple, the cup, and the book. At the top, a progress bar shows the count (1/5), and there is a Pause button. A banner with an apple icon asks, "Can you find the apple?", and a side panel says, "Looking at the apple... Keep your gaze on the object." Nour looks at the apple on the screen, and a progress bar fills while she keeps looking. When it is full, after roughly 1.5 seconds (a proposed design value, not a tested result), the box turns green and a soft chime plays. A "Correct Answer" panel says, "Great Job! You found the apple!", and Nour looks at the Next object button.

The second question asks for the book. Nour glances at the apple first. The progress starts to fill, but she looks away and it quietly resets, with no sound and no penalty. She then looks at the cup by mistake, and this time the selection completes. The "Incorrect Answer" panel says, "Almost There! Let's try finding the book again!" Nour looks at the Try again button, and the same question returns, with the book's box pulsing gently as a hint (a proposed addition). She finds the book. Because she needed a second attempt, this question counts as completed but not as a first-attempt success. The remaining three questions ask for the cup, the apple, and the book again, and Nour finds each of them on the first try.

After the fifth question, the Results screen says, "You Did It! Great job completing the activity!" The illustrative Results design celebrates 5/5 questions completed; Nour's four first-try successes out of five can be recorded separately in a later implementation. The "Your Progress" card says, "Great Learner!" Two large buttons offer Play Again and Back to Home. Nour chooses Back to Home, and her mother closes the session. In a later session, Nour could choose the Hand Gestures card, which uses the same questions, objects, layout, and feedback.

1.2 Five-question session rules

These rules are design proposals for team review. They keep the session short and the task the same in both interaction modes.

- Number of questions: five per session. The header of the game screen shows a progress bar with the count (for example 1/5) and a Pause button.

- How questions progress: the child moves on only after finding the requested object and selecting "Next object". A wrong choice never ends the question, skips it, or stops the session. The "Incorrect Answer" panel offers "Try again", retries are unlimited, and the correct box pulses gently as a hint (a proposed addition).
- Object reuse: the same three objects (apple, cup, book) stay on the table for the whole session and are reused. Each object is asked at least once, two objects are asked twice, and the same object is never asked twice in a row. In Nour's session the order is apple, book, cup, apple, book.
- Effect of incorrect answers: a wrong choice does not lower the number of completed questions. It only means that the question was not answered correctly on the first attempt.
- Results: completing all five prompts produces “Questions completed: 5/5.” The proposed study may separately record first-attempt correctness (e.g., 4/5). An “Objects Found 3/5” figure shown in the current Figma is illustrative and should not be treated as verified accuracy or as a completed-session count.

1.3 Eye Tracking scenario

In this mode, the child selects an object by keeping their gaze on its image in the on-screen camera view. The webcam estimates gaze coordinates on the screen; these are matched to the object boxes already drawn in the live feed.

Step	Child action	Expected system response
1	Nour looks at the Eye Tracking card ("Look at an object to select it") on the "How Would You Like to Play?" screen. Her mother, who pressed START LEARNING on the Welcome screen, confirms the	The Eye Tracking card is highlighted and the eye-tracking setup screen opens.
2	Nour reads or listens to the short instruction.	The "Let's Get Ready!" screen says, "Look at the dots to set up eye tracking." The first target appears with the label "Look at the target". The voice reads the text aloud (proposed).
3	Nour looks at each target as it appears, five times in total.	The system records her gaze at each target. The label shows "Step 1 of 5" and so on, and a progress bar fills. At the end, a green message says, "Calibration Complete! Great job! Your eyes are ready to pla
4	Nour looks at the camera preview, then at START GAME.	The "Let's Get Things Ready!" screen shows the camera preview of the table, a "Camera Ready" check, and three tips under "Before We Start". The system detects and tracks the objects. When "Everything Looks
5	Nour listens and looks at the prompt.	The game screen shows "Find the Object", a progress bar with the count (1/5), and a Pause button. A banner with an apple icon asks, "Can you find the apple?" The prompt stays visible during the whole quest
6	Nour looks at the image of the apple in the live view.	The system compares her gaze position with the on-screen boxes. The box she looks at gets a light highlight, and a side panel says, "Looking at the apple... Keep your gaze on the object."
7	Nour keeps looking at the apple box.	A progress bar or ring fills gradually while her gaze stays on the box. If she looks away, it quietly empties and nothing is selected.
8	Nour keeps her gaze steady until the progress is complete.	After about 1.5 seconds (proposed), the selection is made. The box flashes and a short sound confirms that the choice was registered.
9	Nour waits for a moment.	The system checks whether the selected box is the requested object (apple).
10	Nour sees and hears the result, then looks at the button on the panel.	Correct: the "Correct Answer" panel shows a green check and "Great Job! You found the apple!" with a Next object button. Incorrect: the "Incorrect Answer" panel says, "Almost There! Let's try finding the a

1.4 Hand Gesture scenario

In this mode, the child moves an on-screen pointer with their hand and selects an object by pinching while the pointer is over its box. The tasks, objects, layout, and feedback are the same as in the Eye Tracking mode.

Step	Child action	Expected system response
1	Nour points at the Hand Gestures card ("Point and pinch to select") on the "How Would You Like to Play?" screen. Her mother, who pressed START LEARNING on the Welcome screen, confirms the ch	The Hand Gestures card is highlighted and the hand tutorial screen opens.
2	Nour reads or listens to the three simple steps.	The "Let's Learn the Hand Gesture!" screen says, "Follow these simple steps" and lists "1. Show Your Hand", "2. Move Your Hand" (move your hand to guide the pointer), and "3. Pinch to Choose" (bring your t
3	Nour holds her hand in front of the camera, moves the pointer to TRY IT!, and pinches.	The system confirms that the hand is visible with a green "Hand Detected" message: "Your hand is ready to play!" If the hand is not detected, the message stays inactive (see the recovery section). The pinc
4	Nour looks at the camera preview, then moves the pointer to START GAME and pinches.	The same screens and detection as in the Eye Tracking mode: the camera preview, the "Camera Ready" check, the three "Before We Start" tips, and "Everything Looks Good!" On the game screen, boxes appear aro
5	Nour listens and looks at the prompt.	The same game screen appears: "Find the Object", the progress bar (1/5), the Pause button, and the banner "Can you find the apple?" A side panel says, "Move your hand to the apple and pinch to select!"
6	Nour moves her hand toward the apple on the screen.	The pointer follows her hand across the live view. Any box under the pointer gets a light highlight, so she sees what she would pick.
7	Nour points at the apple box and pinches her thumb and index finger.	The pinch is detected by the child-facing camera. The pointer shrinks slightly to show that the pinch was noticed.
8	Nour keeps the pinch for a short moment.	A pinch counts only when the pointer is over a box. A very short hold (a proposed value to be tested) helps avoid accidental selections. The box then flashes and a short sound confirms the choice.
9	Nour waits for a moment.	The system checks whether the selected box is the requested object (apple).
10	Nour sees and hears the result, then selects the button on the panel with a pinch.	The same feedback panels as in the Eye Tracking mode: "Correct Answer" with Next object, or "Incorrect Answer" with Try again.

1.5 Incorrect answers and error recovery

Suppose Nour is asked to find the book but chooses the cup. The game avoids harsh sounds and uses encouraging wording. The "Incorrect Answer" panel says, "Almost There! Let's try finding the book again!" with a Try again button. Nour selects it, the cup's box returns to normal, and the box around the book pulses gently as a hint (a proposed addition). Nour can try as many times as she needs, and the question ends only when she finds the book. The question then counts as completed, but not as a first-attempt success. If she stops responding for a while (about ten seconds, a proposed value), the voice repeats the prompt and the hint pulses again. The table below shows how the game reacts to other problems. The tag under each situation shows whether the state already exists in the team's Figma. In every case, the message is short, kind, and describes what the game needs, never what the child did wrong.

Situation	Recovery response	HCI rationale
Wrong object selected / [in Figma]	The child selects Try again and the same question returns. The correct box pulses gently as a hint (proposed). Retries are unlimited and there is no penalty or harsh sound.	Feedback; error recovery
Hand tracking lost (Hand Gestures) / [in Figma]	The question pauses. When the hand is visible again, the child selects Try again with the pointer and the same question continues without losing progress.	Visibility of system status
Gaze tracking lost (Eye Tracking) / [in Figma]	Selection pauses. When the gaze returns, the child looks at Try again and the question continues. If it happens repeatedly, the game offers a short re-calibration: "Let's try looking at the dots again." (proposed; no Figma screen	Visibility of system status; error recovery

Situation	Recovery response	HCI rationale
Object disappears from the camera feed / [not in Figma yet]	The question pauses. If the object returns, its box reappears and the question continues. If it does not return after a short wait (proposed, about ten seconds), the game picks another visible object. The "Before We Start" tips (k	Error prevention; recovery
Camera permission denied or camera unavailable / [in Figma]	The adult checks the camera and presses CHECK CAMERA. Nothing starts until the camera works, and the child is never blamed.	Recognition over recall; error recovery
Gaze calibration fails / [not in Figma yet]	After two failed attempts, a grown-up help message suggests sitting closer or facing the light, or using the Back link to return to "How Would You Like to Play?" and choose Hand Gestures.	Error recovery; cognitive load

Note: the grown-up steps (such as CHECK CAMERA) are only for the adult during setup or recovery. The child's own interaction with the game stays based on gaze or hand gestures.

1.6 User journey and hypothetical empathy pathway

The table follows Nour from the Welcome screen to the Results screen. Where the two techniques differ, both are listed. Following the empathy-pathway idea from Lecture 3, the last column adds what Nour might think and feel at each stage. These thoughts and feelings are hypothetical design assumptions, not observed behaviour.

Journey stage	Child action & response	Possible difficulty / proposed support	Anticipated feeling (hypothetical)
1. Welcome screen (START LEARNING)	Sits at the desk while a parent allows the camera, places the objects, and presses START LEARNING. Greets the child and checks camera access.	The child is unsure what to do. Camera access is blocked. One-time adult setup. The "Camera Unavailable" panel (CHECK CAMERA) appears if the camera cannot be used.	Thinks: "What is this game?" / Feels: curious and a little impatient to start.
2. Interaction Mode Selection	Chooses the Hand Gestures card or the Eye Tracking card (the adult confirms). Highlights the card and opens the matching setup screen.	The child does not know which card to pick. Two picture cards with icons and one short line each (recognition over recall). Only two options. A Back link on the next screens.	Thinks: "Which one do I want?" / Feels: excited to choose, but may worry about picking the "wrong" one.
3. Tutorial / calibration	Hand Gestures: reads the three steps, shows the hand, and pinches TRY IT! / Eye Tracking: looks at five targets, then at CONTINUE. Hand Gestures: detects the hand and the pinch. / Eye Tracking: r	The hand is out of view. The child moves their head or looks away. Calibration fails. Short steps. Visible status and progress. A kind retry message. Adult help after two failed attempts. Back li	Thinks: "How do I move the pointer?" or "Why do I look at the dots?" / Feels: focused at first, restless or unsure
4. Camera & Object Setup: camera preview	Waits briefly and looks at the screen. Shows the live preview of the table and checks that the camera works.	The table is too dark, or the objects are outside the view or too close together. The three "Before We Start" tips: place the objects where the camera can see them, keep them apart, and light the	Thinks: "Is that my table on the screen?" / Feels: curious; may feel bored if the wait is long.
5. Camera & Object Setup: object detection	Looks at the screen, then selects START GAME. Detects and tracks the objects and shows when the setup is good.	An object is missed, mislabelled, or the box flickers. The model may not support every object. Large, stable boxes. Start with a small set of familiar, distinct objects (apple, cup, book) and tes	Thinks: "That's my apple!" / Feels: pleased to see real objects on screen; confused if a box looks wrong.
6. Game prompt	Listens and looks at the banner. Asks the question and shows the target picture.	The child forgets what they are looking for. The banner stays visible during the whole question (recognition over recall). The voice repeats after a pause.	Thinks: "I have to find the apple." / Feels: motivated; may feel pressure if the prompt is unclear.
7. Child selection	Eye Tracking: looks at the target. / Hand Gestures: moves the pointer and pinches. Tracks gaze or the pointer and shows progress or hover.	Accidental selection. Noisy tracking. Gaze drifts or the arm gets tired. Large boxes. A dwell of about 1.5 s (proposed). The progress resets quietly. A pinch counts only over a box.	Thinks: "Is it working?" / Feels: concentrated; tired eyes or arm; worried if the progress keeps resetting.
8. Decision and feedback	Watches and listens to the result, then selects the button on the panel. Checks the answer and shows the matching panel.	The child feels that a mistake is their fault or does not understand the result. The same panel format every time, kind wording, and no harsh sounds.	Thinks: "Did I get it right?" / Feels: proud after a correct answer; possibly disappointed

Journey stage	Child action & response	Possible difficulty / proposed support	Anticipated feeling (hypothetical)
			after a wrong one.
9. Retry / next question	Selects Try again or Next object. Keeps the same question or loads the next one. Updates the progress bar.	Frustration after repeated errors. Tracking lost or a camera problem during play. Unlimited retries, a hint, and no penalties. The "Tracking Lost" and "Camera Unavailable" panels explain the prob	Thinks: "I'll try the book." / Feels: determined; may become frustrated after several misses.
10. Session results	Looks at the results and chooses Play Again or Back to Home. Shows the completion message and the first-attempt count, then waits for the choice.	The child does not know how to end, or wants to stop. Two large buttons selected with the same technique. The adult can close the app. No accounts.	Thinks: "I did it!" / Feels: proud and satisfied; may want to play again or may be tired.

1.7 Application flowcharts

The flowchart is split into three connected figures. The Hand Gestures and Eye Tracking branches stay separate (Figure 1) until Camera & Object Setup. Red boxes are error-recovery paths, and the circles A, B, and C connect the figures.

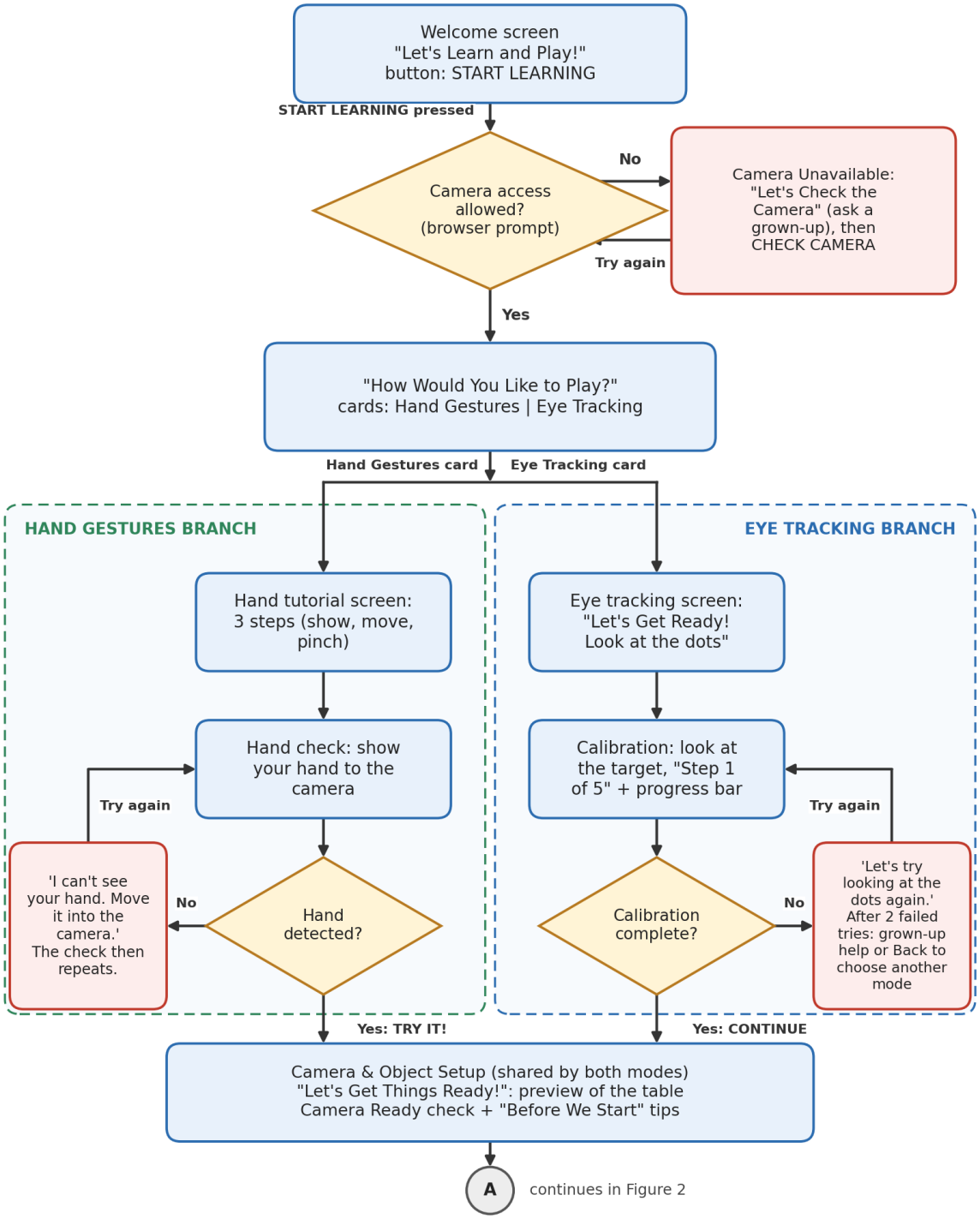


Figure 1. Independent tutorial/calibration paths rejoin at camera setup (Mostafa).

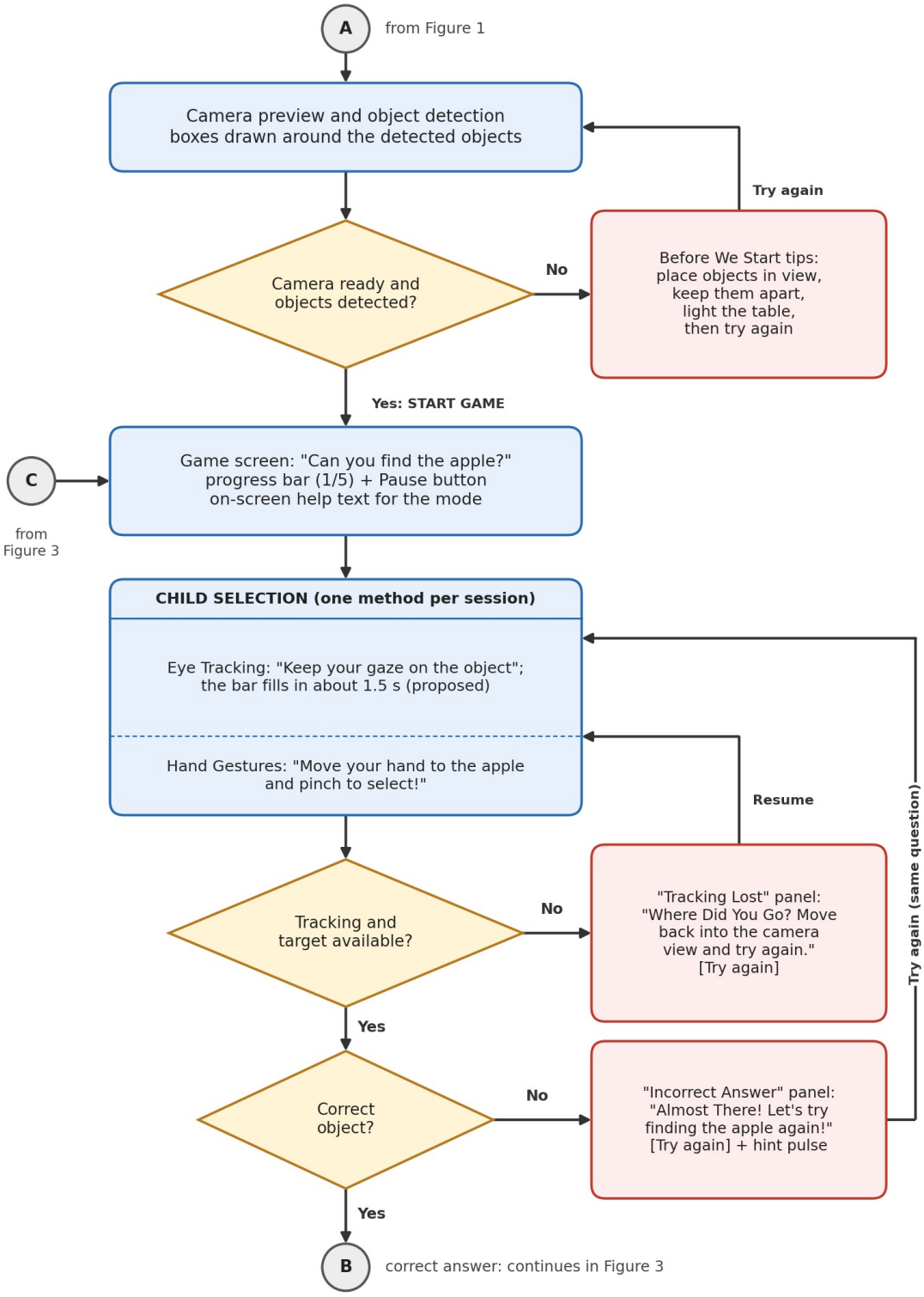


Figure 2. Selection, tracking validation and retry loop (Mostafa).

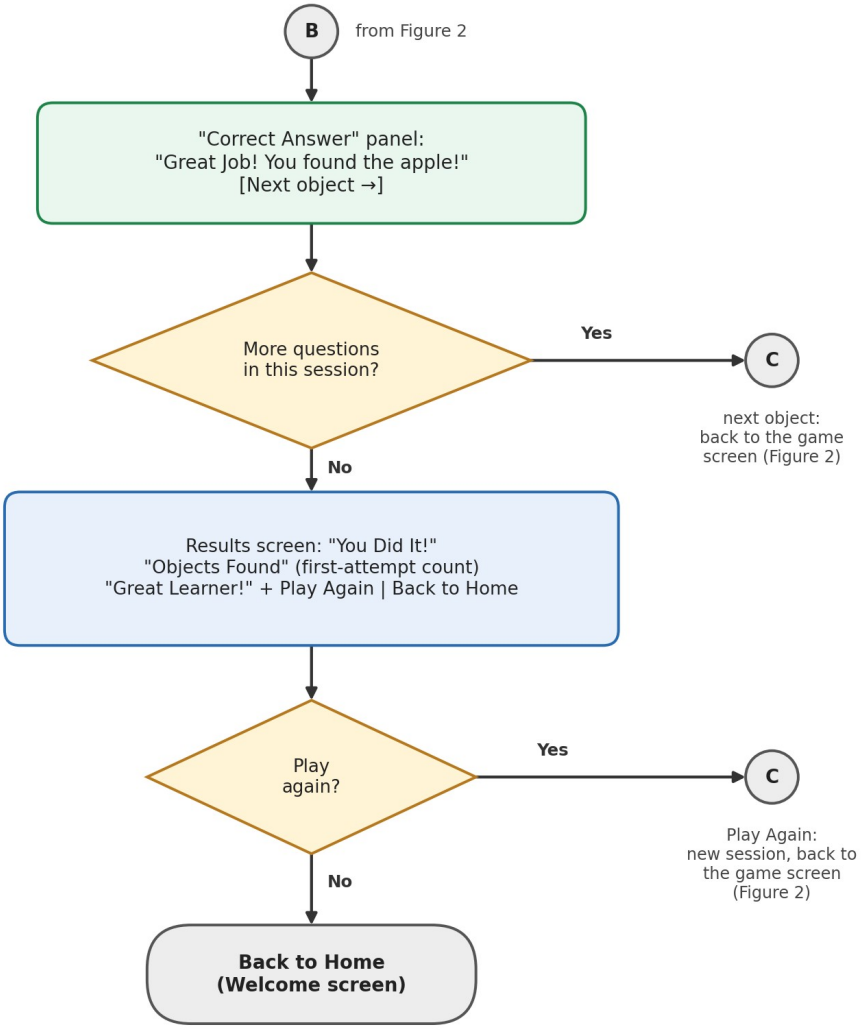


Figure 3. Feedback, question progression and session results (Mostafa).

How to read the flowcharts

- Colours: blue boxes are screens or steps, yellow diamonds are decisions, red boxes are error-recovery messages, the green box is positive feedback, and grey shapes are connectors or the end.
- Connectors: A leads from Figure 1 to Figure 2, B leads from Figure 2 to Figure 3, and C returns to the game screen in Figure 2 for the next object or a new session.
- Child selection: the child uses only the method chosen on the mode screen. The question, the objects, and the feedback panels are the same in both modes.

1.8 HCI analysis of the scenario

Gulf of Execution. The child should always know how to act. The banner shows what to find, the boxes show what can be chosen, and the progress bar or the pointer shows how the choice is made. Short instructions and status messages teach the interaction before the real questions begin: the three-step hand tutorial with its "Hand Detected" message, and the "Step 1 of 5" calibration for Eye Tracking.

Gulf of Evaluation. The child should always know whether the action worked. The filling progress bar, the highlighted box, the flash and sound at the moment of selection, and then the Correct Answer or Incorrect Answer panel give feedback within a moment and in several channels (visual and audio), which follows the multi-modal idea from Lecture 2. This also supports the "visibility of system status" heuristic.

Error prevention and recovery. The dwell time and the pinch-over-a-box rule reduce accidental input, and the quietly resetting progress costs the child nothing. A wrong answer shows an encouraging panel with a Try again button, and tracking problems show the Tracking Lost panel instead of ending the question.

Cognitive load. Instructions are short, each question shows only a few objects (three in the example, following Hick's Law), and the targets are large (Fitts's Law, and the small foveal area explained in Lecture 2). This keeps extrinsic load low, so the child's effort goes to the learning task itself.

Recognition over recall. Picture cards, the banner with an icon, labelled boxes, and the progress bar mean that the child never has to remember a command or what to find.

Empathy pathway (Lecture 3). The user journey table follows the idea of mapping what a user does alongside what they might think and feel at each milestone. The thoughts and feelings listed for Nour are hypothetical design assumptions about a seven-year-old, not observed behaviour. They help the team anticipate friction, such as waiting during setup or frustration after repeated misses, and they should be checked in later user testing.

Taken together, the scenario shows a believable path in which a young child can start, play, make mistakes, recover, and finish with minimal adult help, and in which both interaction techniques use the same tasks and feedback so they can later be compared fairly. The thresholds, the camera set-up, and the wording of proposed items are design proposals. They have not been tested with children, and they will be checked in later phases.

1.9 Evidence limitations

Nour and her family are fictional. No real children were interviewed or observed, and the application has not been implemented. The scenario describes the intended interaction for Phase I. On-screen wording follows the team's Figma. The example objects, the camera set-up, the thresholds, the spoken prompts, and Nour's anticipated thoughts and feelings are design proposals or hypotheses, and they have not been tested.

2. PACT Analysis

Section author: Malak Gamal (247419)

2.1 Purpose and scope

PACT (People, Activities, Contexts and Technologies) is used to identify the user and situation factors that should guide interface requirements (HCI Lecture 3, pp. 14–24). Here it is applied to the proposed "Find the Object" game for children aged 5–10, which compares gaze-dwell selection with a hand-controlled pointer and pinch selection. The two techniques are intended to share the same objects, questions and feedback. The observations below are design assumptions informed by course concepts and the team's Figma screens—not findings from interviews, observations or usability tests.

2.2 People — who will use it?

Physical characteristics. Children differ in visual acuity, hearing, stature, motor control and hand–eye coordination. A younger child may find sustained gaze or a precise thumb–index pinch tiring; others may encounter accessibility barriers in either mode. Target regions and timing should therefore be forgiving and adjustable after testing.

Psychological and cognitive characteristics. Attention, working memory, spatial understanding and frustration tolerance vary between children. A single recognizable object prompt, short steps, clear progress and supportive retries may reduce unnecessary cognitive effort; this is a rationale to test, not a demonstrated benefit.

Knowledge and experience. Children who already use educational games may understand visual prompts quickly, but gaze calibration and air-pinch selection can still be unfamiliar. Reading and language skills also vary, so concise text should be paired with recognizable illustrations; spoken guidance is planned but not yet implemented in Figma.

Frequency and supporting users. First-time users may need full coaching and mode-specific practice, while returning users may prefer fewer explanations. Parents, guardians or teachers are secondary users who can arrange objects, approve camera permissions and help when equipment or tracking fails. The child remains the primary player.

2.3 Activities — what will users do?

The intended sequence is to start the application, select Eye Tracking or Hand Gestures, follow calibration or a brief gesture tutorial, prepare visible objects, respond to an object-finding prompt, receive feedback and complete or repeat a five-question session. The five-question structure is proposed; the Figma result values are illustrative.

Temporal characteristics. Play consists of short repeated selections rather than a continuous precision task. The approximately 1.5-second gaze dwell is an untested starting point; total task duration and ideal break frequency are not known.

Cooperation and complexity. An adult may initially manage the cameras and permission prompts; gameplay should be child-directed. Recognizing the object is conceptually simple, but camera alignment, calibration and selecting a moving or overlapping on-screen region add complexity.

Safety and error recovery. Wrong-object choices should keep the question available for a gentle retry. Tracking loss, unavailable cameras or ambiguous objects should suspend input rather than count as child mistakes. Repeated hand movement, gaze holding and interruptions may cause fatigue or frustration and need later evaluation.

2.4 Contexts — where will use happen?

Physical. Home and classroom spaces differ in light, glare, reflections, available desk space, background clutter and noise. Screen distance, child-facing camera position and arrangement of real objects affect sensing and comprehension. The setup should guide users to adequate lighting, visible separated objects and suitable positioning; audio must not be the only channel in noisy rooms.

Social. A child may play individually with occasional parent help or during a teacher-led activity. Assistance should be available for calibration and failures without taking control of the learning task. Camera use may affect comfort, especially with other children present; the child should understand when sensing is active.

Organizational. Home routines and lesson schedules limit time for repeated calibration. Equipment availability and school rules influence whether the proposed two-camera configuration can be used. Any future recordings or child study would require suitable supervision, guardian permission and responsible data handling, rather than assuming routine storage of faces or gaze data.

2.5 Technologies — what enables interaction?

Input. The proposed object-facing camera supplies the physical-object video, while a separate child-facing camera supplies facial/eye features in gaze mode or hand landmarks in gesture mode. A tracking process would associate detected objects with on-screen boxes. Gaze coordinates would select a box within the displayed video—not locate an object directly in 3D room space. In gesture mode, pointer position plus a deliberate pinch would confirm a selection.

Output. The Figma design depicts the target question, example object display, object outlines, gaze/pointer cues, question progress, readiness messages and four error/answer feedback states. These are illustrated interface states. Spoken prompts and audio responses are intended additions, not demonstrated functional prototype outputs.

Communication and data. A future pipeline would pass camera frames to sensing components, match an eligible tracked target, evaluate the selection and update the screen. Delays, camera synchronization, variable lighting and detection confidence could affect response. No network requirement, processing location or finalized video-storage design is established; minimizing identifiable video retention is an appropriate design constraint.

Data content and limitations. The interface needs supported object names, the active prompt, selected technique, object/track identifiers, tracking status, question progress and recovery messages. Correctness and performance statistics must only appear as actual measurements after the system records them. Gaze accuracy, object overlap, occlusion, pinch thresholds and two-camera feasibility remain unverified.

2.6 PACT summary matrix

PACT area	Key factor for this project	Resulting design direction
People	Ages 5–10; varied reading, attention and motor abilities; adult support.	Short illustrated instructions, clear targets, guided practice and optional assistance.
Activities	Repeated object choices in two modes; accidental selections and fatigue possible.	Comparable five-question task, stable target confirmation, retry without penalty.
Contexts	Home/classroom differences in lighting, space, noise and supervision.	Camera/object readiness guidance, accessible visual feedback, privacy-aware setup.
Technologies	Two proposed camera streams, tracking, gaze mapping and pinch recognition.	Show system status; block uncertain input; validate feasibility before claims.

2.7 Provisional child personas

The following fictional personas illustrate possible user differences across the target age range. They are hypotheses for design discussion, not profiles derived from interviews, ethnography or actual participants.

Lina, age 6 | Early-reader / home learning / Context and experience: Plays short picture-based learning games at home with a parent; recognizes everyday fruits and toys but reads slowly. / Abilities and needs: Can follow one-step visual demonstrations; precise pinching and prolonged gaze may be difficult on some attempts. May need adult help with camera setup. / Goal and motivation: Wants to find familiar objects and receive encouraging confirmation. Long instructions or unexplained failures may be discouraging. / Design implication: Keep each prompt short, make practice repeatable, use clear object images and visual feedback; add spoken guidance if implemented.

Omar, age 9 | Confident learner / classroom use / Context and experience: Uses school learning apps and understands brief written instructions; may play in a supervised classroom setting. / Abilities and needs: May manage the basic game independently but is still new to webcam gaze calibration and pinch recognition. Background activity can interrupt attention. / Goal and motivation: Wants to complete the game at a comfortable pace and understand whether his selection was registered. Repeated setup after failures may frustrate him. / Design implication: Provide visible input-status feedback and progress, allow calm retries, and reduce unnecessary steps for returning users after validation.

2.8 Design requirements derived from PACT

These requirements are recommendations derived from the analysis. Some are represented visually in Figma; sensing, audio and data safeguards need future implementation and testing.

ID	Proposed requirement	PACT rationale
R1	Use short object prompts with large readable	People: reading ability and attention vary; Contexts: audio

ID	Proposed requirement	PACT rationale
	text and pictures; offer spoken prompts when developed.	may be missed in class.
R2	Offer two clearly named input modes with a simple explanation of each.	People: prior experience varies; Activities: an understandable initial choice.
R3	Provide a three-step hand tutorial or step-by-step gaze calibration, with clear retry guidance.	People: unfamiliar controls; Activities: setup and learning burden.
R4	Guide camera permissions, object spacing and lighting before play; let adults assist.	Contexts: different rooms and supervision; Technologies: camera dependence.
R5	Register choices only for stable, eligible tracked targets; avoid duplicate pinch and unreliable dwell.	Activities: prevent accidental input; Technologies: uncertainty and latency.
R6	Keep the object task, prompt count, layout and response style comparable across both modes.	Activities: consistent learning demand; People: reduced relearning.
R7	Give supportive correct/incorrect feedback and recovery guidance for tracking or camera failures.	People: confidence and frustration; Activities: safe error recovery.
R8	Present session progress clearly; show accuracy only when measured and minimize child data collection.	Technologies: reliable results; Contexts: privacy and adult oversight.

2.9 PACT-to-Figma mapping

This mapping uses the screen evidence reproduced in Ahmed Wahba's application-design report. "Shown" refers to mockup content or simulated navigation only. Requirements describing real sensing, audio or data handling remain proposals; they are not operational Figma functions. Table codes: P = People, A = Activities, C = Contexts and T = Technologies.

PACT finding	User need	Design requirement / evidence status	Figma screen group
Limited reading and attention (P)	An easy, inviting starting point.	R1: concise message, illustration and prominent Start Learning; shown. Audio is only planned.	01 — Welcome
Different prior experience (P/A)	Understand two modes without technical terms.	R2: illustrated Hand Gestures / Eye Tracking choices with brief descriptions; shown.	02 — Interaction Mode Selection
Unfamiliar pinch actions (P/A/T)	Practice before answering.	R3: show-hand, move-pointer and pinch steps, plus Hand Detected state; shown as design, not detection.	03 — Hand Gesture Tutorial
Gaze requires calibration (A/T)	Know where to look and when ready.	R3: visible dots, step progress, completion state; shown as design, not real calibration.	04 — Eye Tracking Calibration
Lighting and spacing vary (C/T)	Prepare camera and objects safely.	R4: camera preview example, object-separation guidance and readiness state; shown only.	05 — Camera & Object Setup
Selection can be uncertain (A/T)	Focus on target and know input status.	R5–R6: same five-question format in both variants, differing pointer/gaze cues; shown. Stable target gating is proposed.	06 — Main Gameplay (two variants)
Mistakes and sensor failures (P/A/T)	Retry without blame or losing the task.	R7: correct, incorrect, tracking-lost and camera-unavailable panels; shown. Automatic recovery remains proposed.	07 — Feedback States
Users need meaningful closure (P/A/T)	See completion without misleading scores.	R8: Play Again / Back to Home and progress illustration; shown. The 3/5 mockup is illustrative, not validated accuracy.	08 — Session Results

2.10 Conclusion and evidence boundary

The PACT analysis identifies child-development differences, repeated task demands, variable environments and sensor uncertainty as the main requirements drivers. The team's current Figma prototype represents the eight screen groups and the intended flow, while the proposed two-camera detection, gaze-dwell and pinch mechanisms still require feasibility checks, implementation and supervised evaluation. Accordingly, this section offers requirements and provisional personas rather than invented user observations or measured benefits.

3. Survey & Critical Analysis

Section author: Mohamed Ramadan (246813)

3.1 Survey Overview

Our project, Find the Object, is a camera-based learning game for children aged 5 to 10. The game shows a live camera view of everyday objects and asks the child to find one of them. We will compare two touch-free selection techniques: eye tracking (the child looks at an object for about 1.5 seconds) and hand gestures (the child moves an on-screen pointer and pinches to select). Because the system is still at the design stage, we prepared a short online questionnaire for parents and teachers. Its role is to collect information about the target users, their current difficulties, their learning environment and their expectations before we design and build the system.

The survey has 12 closed-ended questions and one optional open-ended question, and it takes about 5 to 7 minutes to complete. It is ready to be published with Google Forms (Appendix A). At the time of writing the survey has not been distributed, so this section presents the survey design and its critical analysis, not results.

3.2 Research Objective

Objective: to understand how children aged 5 to 10 currently use educational apps, what problems they face with existing input methods, what equipment and support are available to them, and what parents and teachers expect from a camera-based game that uses eye gaze or hand gestures for selection.

The survey is designed to answer the following questions:

RQ1. How often do children aged 5 to 10 use educational apps, and what difficulties do they currently have with touchscreens, mice or keyboards? (Q3, Q4)

RQ2. Where and on which devices could the game realistically be used? (Q6)

RQ3. Which types of instructions suit this age group? (Q5)

RQ4. How familiar are children with touch-free (movement-based) interaction? (Q7)

RQ5. How difficult do parents and teachers expect gaze-based and gesture-based selection to be? (Q8, Q9)

RQ6. What supervision, accessibility and privacy needs must the design meet? (Q10, Q11, Q12)

Scope: the survey measures anticipated expectations and opinions. It cannot show which technique is faster, more accurate or more enjoyable for children, because nobody has used either method yet. That comparison requires supervised usability testing with children in later phases (see 3.8).

3.3 Target Respondents

The primary respondents are:

Parents or guardians of at least one child aged 5 to 10, who see how the child uses technology at home.

Teachers of children aged 5 to 10, who see many children of the same age and the classroom setting.

These adults are appropriate at this early stage because they can describe children's current habits, difficulties, devices and supervision needs, and they are the people who would allow and set up a camera-based game. Surveying young children directly with an online questionnaire is not practical, because many children aged 5 to 7 cannot read or answer rating scales reliably, and it would also raise consent issues.

However, adult answers are proxy reports. Parents and teachers can only estimate how a child would react to eye or hand control, so their answers cannot replace observing children using the real system. Question 1 screens out respondents who do not belong to the target group, and the description at the start of the form explains both methods in the same neutral way so that respondents are not pushed toward either one.

3.4 Survey Design Rationale

We designed the survey using the guidance from Lab 3 (slides 15 to 20). The table below shows how each guideline was applied.

Lab 3 guideline	How it is applied in our survey
Start with the problem	The research questions (3.2) were written first, and every survey question is linked to one of them. Questions whose answers we would not use were removed.
Identify the target users	Parents and teachers of children aged 5 to 10, with a screening question (Q1) that ends the survey for anyone else.
Questions must be specific	Questions name a clear time period ("in a typical week"), concrete behaviours (e.g. "pressing small buttons") and concrete devices, instead of vague words like "often" or "technology".
Avoid biased questions	No question assumes that the game, eye tracking or gestures are good. Both methods are described with the same structure and level of detail, and Q8 and Q9 ask about expected difficulty using identical wording and scales.
Balanced answer choices	Rating scales have two negative options, a neutral middle and two positive options. Ranges do not overlap (e.g. 5 to 6, 7 to 8, 9 to 10 years). "Not sure" or "None of these" options are included where respondents may not know the answer.
Prefer closed questions	12 of the 13 questions are closed (multiple choice, checkboxes or a grid) so they can be counted and compared. Only the last question is open-ended and optional.
Avoid unexplained terms	No technical terms such as YOLO, MediaPipe, bounding boxes or dwell-time threshold appear in the form. "Eye tracking" is described as "looking at the object for about one and a half seconds".
Be considerate of respondents	The survey is short (about 5 to 7 minutes), anonymous and voluntary, explains why the answers matter, and does not collect names, photos or other identifying information about adults or children.

3.5 The Questionnaire

Form title: Find the Object: Parent and Teacher Survey

Introduction shown to respondents (neutral description of both methods):

In the game, the screen shows a live camera view of everyday objects, such as an apple, a cup or a book. The game asks the child to find one object, for example: "Can you find the apple?" The child chooses an object without using a mouse, keyboard or touchscreen. We are considering two ways of choosing:

Method A (Looking): the child looks at the object on the screen. If the child keeps looking at it for about one and a half seconds, the object is selected.

Method B (Hand movement): the child moves one hand in the air to move a pointer on the screen, then pinches the thumb and index finger together over the object to select it.

Both methods use the same game, the same objects and the same pictures and sounds. If the child chooses the wrong object, the game asks them to try again.

The full introduction, including the purpose, intended respondents and the privacy statement, is given in Appendix A.

Form section: About you

Q1. Which of the following best describes you?

Multiple choice (one answer) · Required

Parent or guardian of a child aged 5 to 10

Teacher of children aged 5 to 10

Both a parent/guardian and a teacher of children aged 5 to 10

None of the above

Branching: Choosing "None of the above" ends the survey (screening question).

Form section: About the child

Q2. Which age group do you mainly have in mind when answering this survey?

If you know children in more than one age group, choose the one you know best. | Multiple choice (one answer) · Required

5 to 6 years

7 to 8 years

9 to 10 years

Q3. In a typical week, on how many days does the child use educational apps or games?

Multiple choice (one answer) · Required

0 days

1 to 2 days

3 to 4 days

5 to 7 days

Not sure

Branching: Choosing "0 days" skips Question 4.

Form section: Current use of educational apps

Q4. When the child uses educational apps on a touchscreen, mouse or keyboard, how often do you notice the following?

Multiple-choice grid (one answer per row) · Required

Q4 response scale

For each of the five behaviors, use one response: Never / Rarely / Sometimes / Often / Always / Not sure.

- Selecting the wrong item by mistake
- Having trouble pressing or clicking small buttons
- Not understanding written instructions
- Needing an adult's help to continue
- Losing interest before finishing an activity

Form section: Learning setting and preferences

Q5. Which types of instructions would most help a child of this age understand what to do in a learning game?

Select up to 3. | Checkboxes, up to 3, with "Other" · Required

Spoken (audio) instructions

Short written instructions

Pictures or icons

A short animated demonstration

An adult explaining the activity

Other (please specify)

Q6. Which of the following devices could the child use to play a learning game that needs a camera?

Select all applicable devices. Choose “None of these” or “Not sure” only on its own. | Checkboxes · Required

A laptop or desktop computer with a camera, at home

A tablet with a camera, at home

A smartphone with a camera, at home

A school computer, laptop or tablet with a camera

None of these

Not sure

Q7. Has the child ever played a game or used an app that is controlled by body, hand or eye movement instead of touching the screen?

For example, a motion-controlled video game. | Multiple choice (one answer) · Required

Yes, often

Yes, once or a few times

No, never

Not sure

Form section: Two ways to choose an object

This section repeats the description of Method A and Method B shown in the introduction.

Q8. Method A (Looking): how easy or difficult do you expect it would be for the child to select an object by looking at it for about one and a half seconds?

Multiple choice (one answer) · Required

Very easy

Somewhat easy

Neither easy nor difficult

Somewhat difficult

Very difficult

Not sure

Q9. Method B (Hand movement): how easy or difficult do you expect it would be for the child to select an object by moving a pointer with their hand and pinching their fingers together?

Multiple choice (one answer) · Required

Very easy

Somewhat easy

Neither easy nor difficult

Somewhat difficult

Very difficult

Not sure

Form section: Support, accessibility and privacy

Q10. During a short game session of about five questions, how much adult help do you expect the child would need?

Multiple choice (one answer) · Required

No help

Help only to set up the camera and start the game

Help a few times during the game

Help throughout the whole game

Not sure

Q11. Which accessibility needs should the design team investigate first to make this game more inclusive for children aged 5 to 10?

Select up to 3 accessibility needs to investigate first. | Checkboxes, up to 3, with “Other” | Required

Compatibility with glasses

Alternatives for limited hand or arm movement

Adjustments for low vision

Flexible timing for difficulty remaining still or focused

Instructions accessible to children learning to read

Support for left-handed gestures

Other (please specify)

Q12. Suppose the game uses a camera to follow where the child is looking or how their hand moves while they play. How comfortable would you feel about this?

Multiple choice (one answer) · Required

Very uncomfortable

Somewhat uncomfortable

Neither comfortable nor uncomfortable

Somewhat comfortable

Very comfortable

Not sure

Q13. Is there anything else you would like us to consider when designing this game? (Optional)

Please do not include any names or personal details. | Paragraph (open-ended) · Optional

[Free-text answer]

3.6 Question-by-Question Justification

The table explains why each question is included, why its wording and answer choices are appropriate, and what limits the answer may have.

Question	Research purpose	Why appropriate	Potential limitation
Q1 Relationship to children	Screens respondents and separates parent and teacher views.	Closed and complete; "None of the above" ends the survey so only relevant people answer. Asks about role only, not identity.	Self-reported and cannot be verified. Parents and teachers see children in different settings, so the two groups may answer differently.
Q2 Age group	Lets us compare expectations for younger and older children, whose motor control, attention and reading skills differ.	Equal two-year ranges with no overlap, avoiding the unequal age ranges shown as a bad example in Lab 3.	Teachers of mixed-age classes must choose one group. Some groups may receive few answers.
Q3 Frequency of app use	Measures how familiar children are with educational apps; also used to skip Q4 for children who never use them.	"In a typical week" and day ranges are more specific than often/sometimes/rarely (Lab 3 tip). Ranges do not overlap and "Not sure" is included.	Based on memory and estimation. Parents may not know about school use, and teachers may not know about home use.

Question	Research purpose	Why appropriate	Potential limitation
Q4 Current difficulties	Identifies real problems with existing input methods, which helps show whether there is a need for a touch-free alternative.	Each row describes one observable behaviour. The frequency scale is balanced and includes "Not sure".	Grids can cause respondents to give the same answer to every row. Finding problems with touchscreens does not prove that gaze or gestures will solve them.
Q5 Instruction formats	Guides how the game gives instructions to children who may not read well yet.	"Up to 3" makes respondents choose priorities instead of selecting everything. Includes adult help and "Other".	Choices are not ranked, so we cannot tell the single most useful format. Language (Arabic or English) is not asked.
Q6 Devices and setting	Shows which devices and settings (home or school) are available, which affects camera placement and screen size.	Concrete devices, each linked to home or school. "None of these" and "Not sure" are included.	Does not capture camera quality, position or whether a second camera is available for the proposed two-camera setup.
Q7 Familiarity with touch-free control	Tells us whether children have used movement-based control before, which helps interpret Q8 and Q9.	A simple example explains the idea without technical terms. Answer choices are balanced and include "Not sure".	Respondents may not recognise every type of movement-based game. "Often" is still somewhat open to interpretation.
Q8 Expected difficulty: Looking	Collects adults' expectations about how hard gaze-based selection would be at each age.	Asks about expected difficulty, not "which is better". Uses exactly the same scale as Q9 so the two can be compared.	Hypothetical: respondents have not seen the method. The 1.5-second detail may be hard to imagine. Method A always appears first (order effect).
Q9 Expected difficulty: Hand movement	Same as Q8 for gesture-based selection.	Parallel to Q8 in wording, length and scale, so neither method is favoured.	Same as Q8. Pinching may be easier to imagine than gaze control, which could affect answers.
Q10 Adult supervision	Shows whether we need a separate adult setup step and whether children can play alone.	Levels are ordered from no help to help throughout, and match the project assumption that adults may help with setup.	Depends on session length, which respondents can only estimate. Parents may overstate their child's independence.
Q11 Accessibility needs	Identifies which inclusive design needs the team should prioritise.	Asks about needs in general, not about a specific child, to protect privacy. "Up to 3" forces priorities.	Shows opinions, not how common each need is. The list may miss some needs ("Other" helps). Some respondents may find the topic sensitive.
Q12 Comfort with camera use	Measures how acceptable camera use is to adults, which affects the privacy design.	Neutral "Suppose..." wording without promises that could bias answers. Balanced five-point scale plus "Not sure".	Comfort may depend on safeguards (e.g. no recording) that are not described. One question cannot show which safeguard matters most.
Q13 Optional comments	Captures ideas or concerns that the closed questions did not cover.	Lab 3 allows a comment field to give room for the respondent's own opinion. It is optional, so it does not add pressure.	Open answers are harder to analyse and must be grouped by hand. Many respondents may skip it.

3.7 Survey Critique and Limitations

3.7.1 Problems found in our draft questions

Before finalising the survey, we reviewed our first draft against the Lab 3 examples of bad survey questions. The main problems and our fixes are shown below.

Draft wording	Problem (Lab 3 principle)	Final version
"Do you think eye tracking will make learning more exciting?"	Leading: assumes a positive benefit and favours one technique. "Eye tracking" may be unfamiliar.	Q8 and Q9: two parallel questions about expected difficulty, using the same scale for both methods.
"Which is better for children, eye tracking or hand gestures?"	Asks respondents to judge performance without having used either method; "better" is vague.	Removed. Expectations are asked separately in Q8 and Q9; the real comparison is left to usability testing.
"What dwell time should the	Technical term that respondents cannot	Removed. The method is described in plain words:

Draft wording	Problem (Lab 3 principle)	Final version
gaze selection use?"	understand or answer.	"keeps looking at it for about one and a half seconds".
"Child's age: 5 to 6 / 7 to 10"	Unequal ranges (the same mistake as the Lab 3 age example).	Q2: equal ranges of 5 to 6, 7 to 8 and 9 to 10 years.
"Does your child use educational apps? Yes / No"	Too broad; answers cannot be grouped by how much the child uses apps.	Q3: number of days in a typical week, with non-overlapping ranges.
"Would you let the game record your child?" Yes / No	"Record" is alarming and not accurate for the design, and a yes/no answer gives no middle ground.	Q12: neutral description of camera use with a balanced comfort scale.

3.7.2 Limitations of the final survey

Even after these changes, the survey has limitations. Each one is described below with why it matters and a realistic way to reduce it.

Limitation	Issue and why it matters	Realistic improvement
Sampling bias	If the link is shared only through our own families and contacts, respondents may come from similar schools, education levels and income groups. The results would then not represent all parents and teachers of children aged 5 to 10.	Share the form through more than one channel (e.g. a school with permission, parent groups and teachers we do not know), and report the number of parents and teachers and the age groups (Q1, Q2) so readers can judge the sample.
Proxy responses	Adults answer for children. They may overestimate or underestimate what a child can do with eye or hand control.	Treat answers as expectations only, and confirm them later by observing children using the prototype under supervision.
Technology unfamiliarity	Many respondents have never seen gaze or gesture control, so they may misunderstand the methods or choose "Not sure" often.	Pilot-test the description with 3 to 5 people. If it is still unclear, add two short videos of equal length showing each method, so neither method looks more attractive.
Hypothetical preference	People may expect one method to be easy, but find it difficult in practice (or the other way around).	Do not use Q8 and Q9 to decide which technique is better. Use them only to plan what to test, and compare techniques using measured task time, accuracy and errors.
Question order	Method A (Looking) always appears before Method B, which may influence how Method B is rated.	Create a second copy of the form with Method B first, and send each version to half of the respondents.
Wording and language	The survey is in English, but many parents in Egypt may prefer Arabic. Unclear wording can change the meaning of answers.	Provide an Arabic version, translated and checked by a native speaker, and pilot-test both versions before sending.
Answer choices	"Not sure" options prevent forced answers, but some respondents may use them to avoid thinking. "Select up to 3" may hide a fourth important choice.	Report "Not sure" answers separately instead of removing them. Use the open comment (Q13) to catch missing choices, and add an "Other" field where needed.
Survey fatigue	13 questions, including a grid, may make some respondents answer quickly or give the same answer to every row.	Keep the survey at about 5 to 7 minutes, show a progress bar, keep only one grid, and check for identical answers across all rows of Q4.
Social desirability	Parents may describe their child as more capable or independent, and teachers may report fewer classroom problems.	Keep the survey anonymous, avoid judgemental wording, and remind respondents that there are no right or wrong answers (already in the introduction).
Privacy	Questions about children and cameras are sensitive. Collecting names or emails could identify a child.	Turn off email collection in Google Forms, ask no names or photos, ask about children in general (Q11), and use the answers only for this project.

3.8 Proposed Improvements

Pilot test: ask 3 to 5 parents and teachers to complete the form and explain any confusing wording before the real distribution.

Two language versions: English and Arabic, checked for equal meaning.

Counterbalanced order: two versions of the form with Method A and Method B presented in opposite orders.

Wider distribution: more than one school or community, with the sample composition reported honestly.

Follow-up usability evaluation: in later phases, a supervised test with children (with parental consent and the child's agreement) should measure task completion time, selection accuracy, number of errors and retries, observed engagement, and satisfaction using a child-friendly scale such as smiley faces. Only this evaluation can show which technique works better for children.

3.9 Conclusion

The survey is designed to give our team an early, structured view of the people who will use and supervise Find the Object, before any system is built. It follows the Lab 3 guidance by starting from clear research questions, targeting the right respondents, using specific and neutral wording, and offering balanced, mostly closed answer choices. The table below shows examples of how possible answers could guide design decisions.

If the responses show that…	Possible design decision
Many adults expect looking (Q8) to be difficult for 5 to 6 year olds	Add a short practice round, show a visible progress ring while the child looks at an object, and test different looking times in the usability study.
Many adults expect hand movement (Q9) to be difficult or tiring	Use large objects and pointer, keep sessions to about five questions, and allow short breaks.
Spoken instructions and pictures are chosen most (Q5)	Give every instruction as audio with an icon, and keep on-screen text to a minimum.
Laptops are the most available device (Q6)	Design first for a laptop with one built-in camera, and re-check whether the proposed two-camera arrangement is practical.
Adults expect help is needed for setup only (Q10)	Create a separate adult setup screen for camera position and permissions, then hand over to the child.
Many respondents are uncomfortable with camera use (Q12)	Process images on the device without saving them, show when the camera is on, ask an adult for permission first, and explain this clearly.

These findings would describe needs and expectations only. Whether eye tracking or hand gestures is faster, more accurate, more engaging or more satisfying for children must be decided later through supervised usability testing of both techniques.

4. Application Design & HCI Principles

Section author: Ahmed Wahba (245559) | Source: reviewed Figma design report

The Figma prototype communicates eight main screen groups for children aged 5–10 and two equivalent gameplay modes. It represents the proposed interface and simulated navigation; it is not a working camera or AI implementation. The primary design objectives are to make instructions understandable, keep layouts consistent, prevent errors, provide feedback, and let children recover gently when a selection or sensor state fails.

4.1 Screen structure and intended behavior

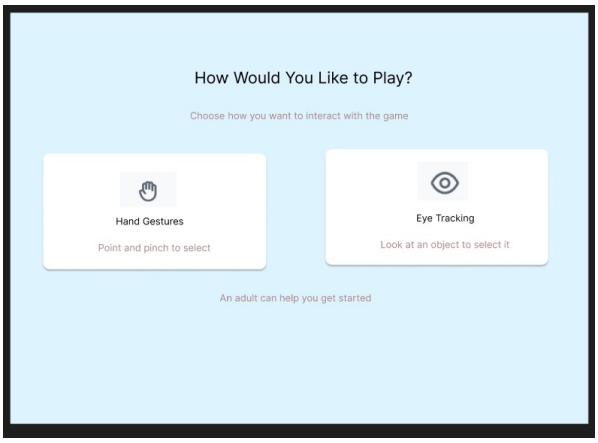
Screen group	Main purpose	Intended interaction / feedback
01 Welcome	Introduce the learning game with child-friendly illustration and one clear next action.	“Let’s Learn and Play!” and Start Learning; adult may help before calibration.
02 Mode Selection	Explain two input choices without technical terminology.	Hand Gestures card vs. Eye Tracking card; each opens its own setup path.
03 Hand Gesture Tutorial	Teach hand positioning, moving a pointer and pinching.	Three illustrated steps and Hand Detected readiness state (mockup only).
04 Eye Tracking Calibration	Prepare gaze mapping before gameplay.	Look at successive dots; visible progress, completion and retry states.
05 Camera & Object Setup	Arrange visible objects and check setup readiness.	Camera preview representation, spacing and lighting guidance, Start Game.
06 Main Gameplay	Ask the child to find an object using a common display across modes.	Five-question format with an object-view panel, highlight, progress and mode-specific gaze/pointer cues.
07 Feedback States	Communicate correct, incorrect and technical-error outcomes.	Great Job / Almost There / Tracking Lost / Camera Unavailable, with recovery actions.
08 Session Results	Provide encouraging completion and choices to replay or return.	“You Did It!”, progress illustration, Play Again and Back to Home.

4.2 Figma screen evidence

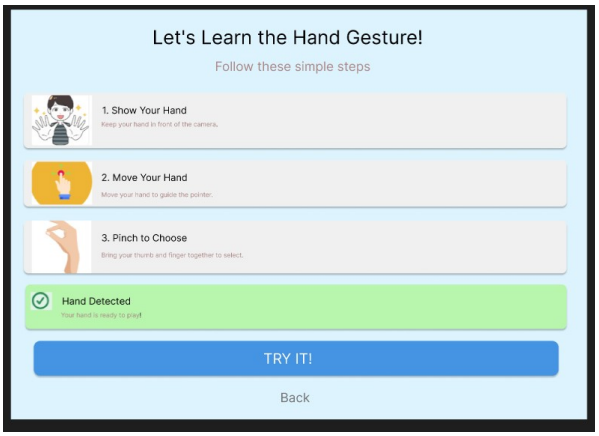
The following figures reproduce the completed screen mockups from Ahmed’s reviewed PDF. Labels and counters inside these images are illustrative prototype data; they do not establish operating detection or measured session performance.



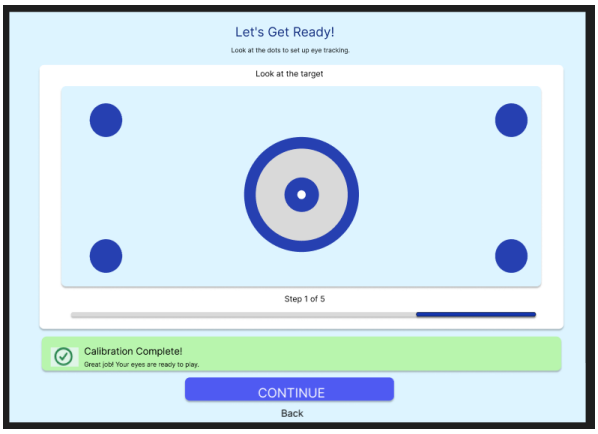
Welcome



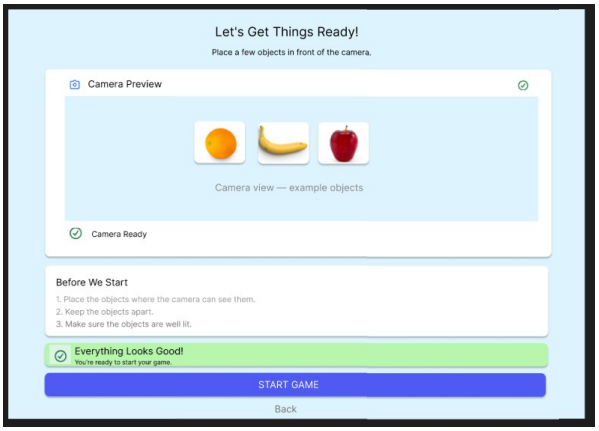
Interaction Mode Selection



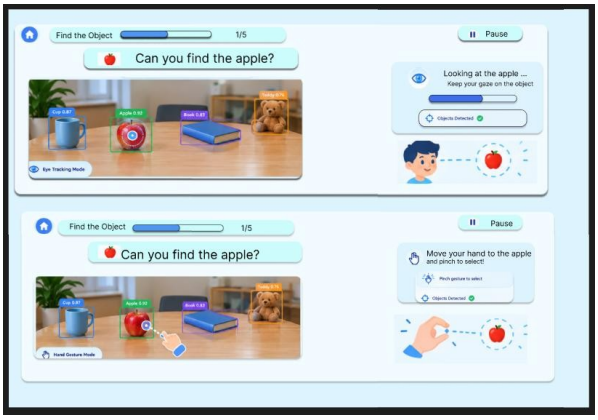
Hand Gesture Tutorial



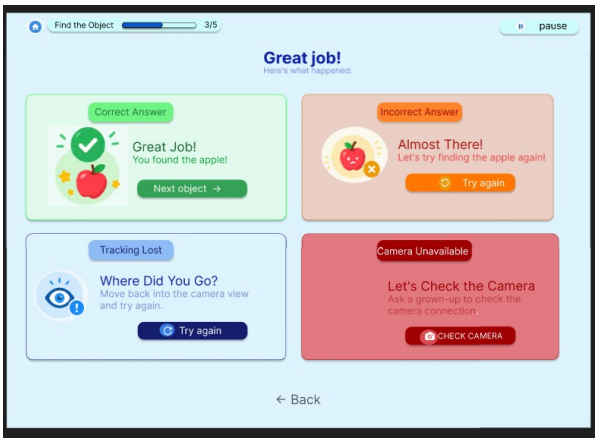
Eye Tracking Calibration



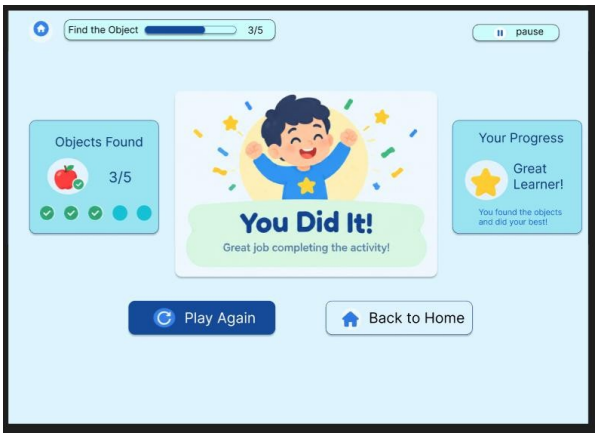
Camera & Object Setup



Main Gameplay: two input variants



Feedback and error states



Session Results

Results-screen clarification: a “3/5” number visible in the Figma mockup is illustrative. In the proposed five-question scenario a completed session is described as 5/5 questions completed; first-attempt correctness would be calculated separately only after implementation.

4.3 Application navigation and recovery

The two interaction paths remain separate until Camera & Object Setup: Welcome → Mode Selection → Gesture Tutorial OR Gaze Calibration → Camera & Object Setup → Gameplay → Feedback → Next Question or Retry → Session Results. Incorrect selection retries the same prompt; missing tracking pauses interaction; unavailable camera or unsuccessful calibration routes to a recoverable setup state.

Figure 1A. Setup and mode preparation

Two independent interaction paths; adult assistance for permission or recovery

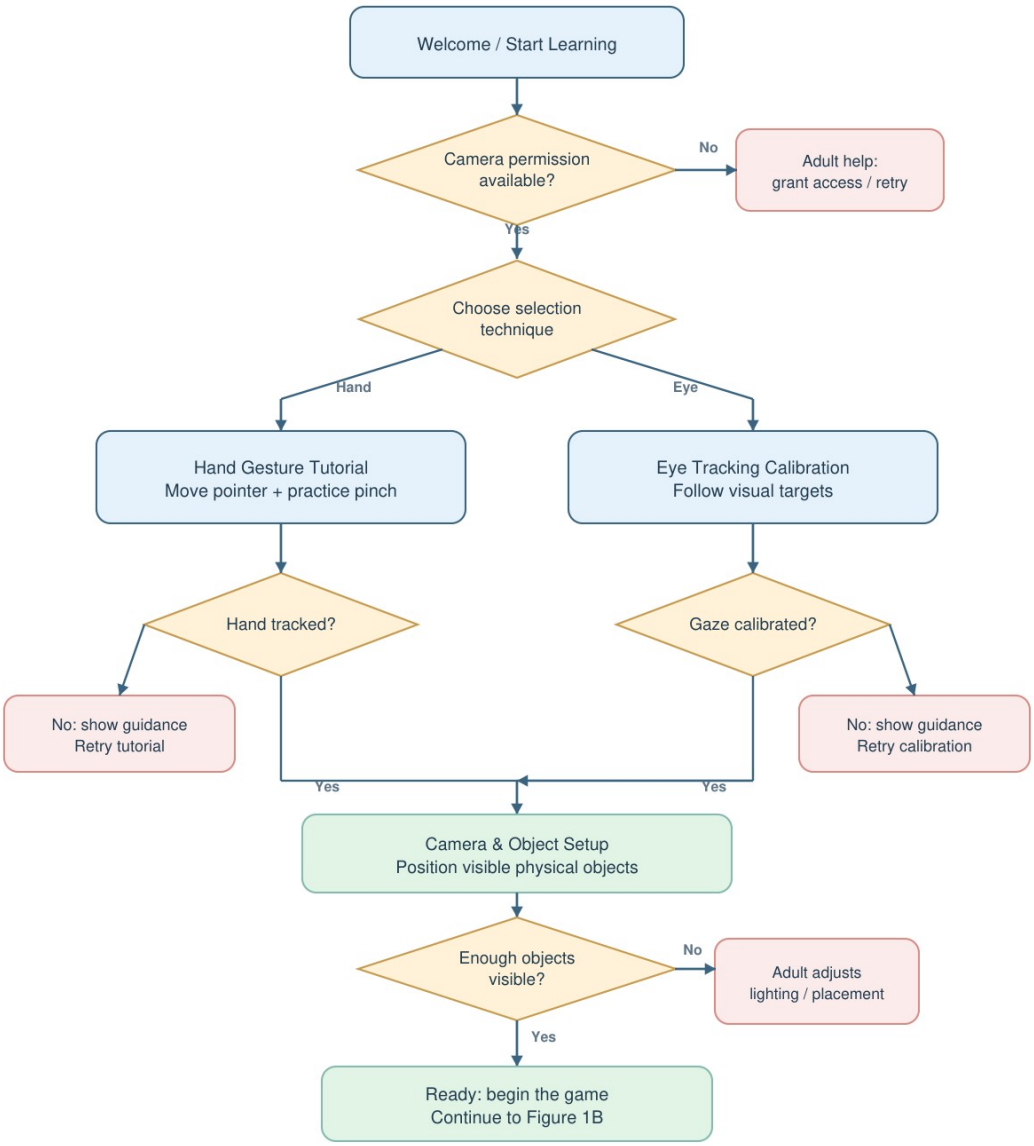


Figure 4. Setup and preparation flow, reproduced from Ahmed's reviewed PDF.

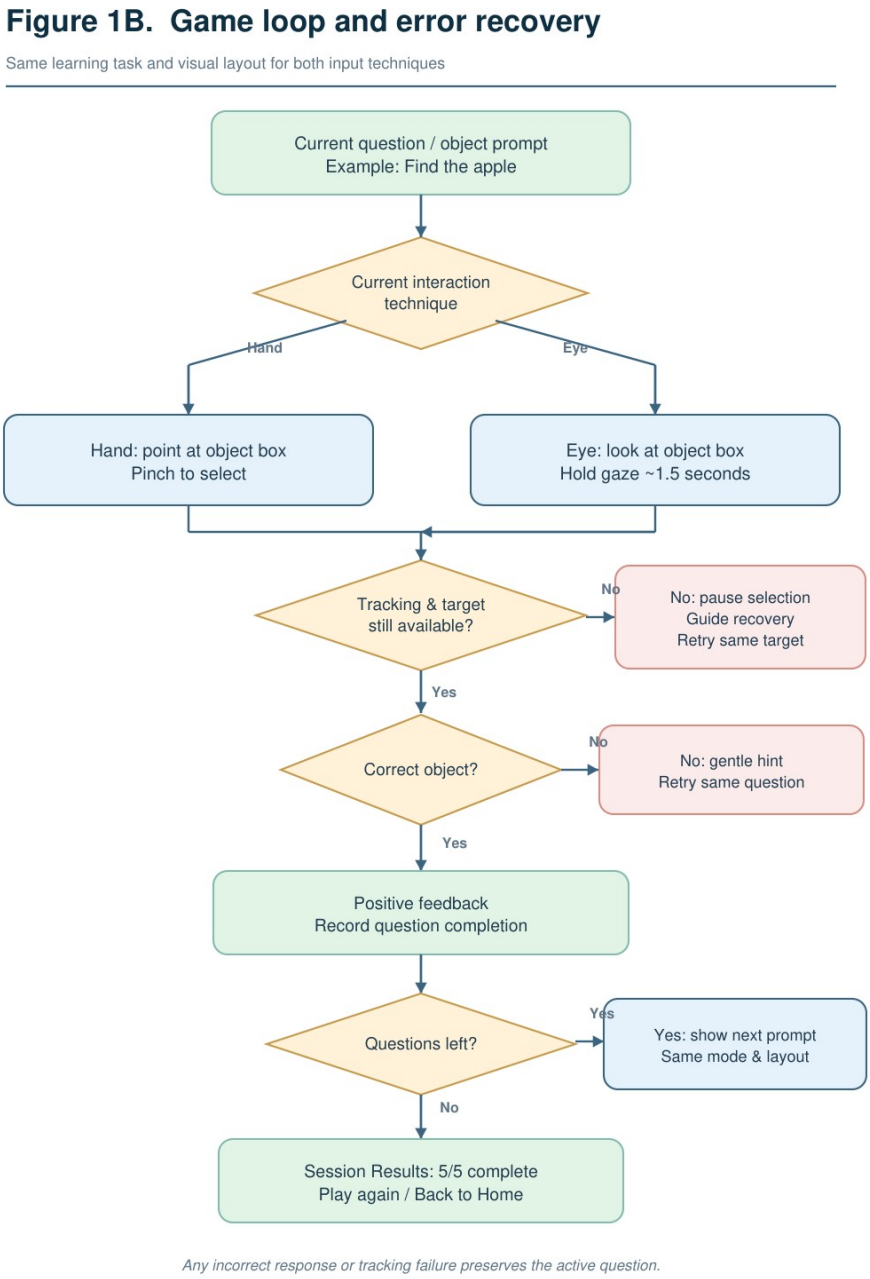


Figure 5. Gameplay and error-recovery flow, reproduced from Ahmed's reviewed PDF.

4.4 Seven Lab 2 HCI principles

Lab 2 principle	Where applied	How represented	Why appropriate
Navigation	Welcome, mode choice, tutorials, recovery and results	A visible next action (Start, Continue, Back, Try Again, Play Again) makes progress apparent.	Children can understand where they are and how to proceed.
Familiarity	Welcome, mode choice and gameplay	Everyday object pictures and a concrete "Find the apple" prompt replace jargon.	Uses familiar visual concepts rather than technical terminology.
Consistency	All screens, both modes	Shared light-blue styling, typography, prompt, object layout and feedback language.	Reduces relearning and supports a fair interaction comparison.
Error Prevention	Tutorial, calibration, camera setup, gameplay	Guided hand/gaze setup, object-spacing tips, and planned block on uncertain	Reduces unintended selection and confusion; real checks remain future

Lab 2 principle	Where applied	How represented	Why appropriate
		selection.	work.
Feedback	Setup, calibration, correct/incorrect/error states	Readiness, progress and recovery messages make system state visible.	Helps users tell whether an action registered or needs correction.
Visual Clarity	Welcome, gameplay, calibration	Brief prompts, visually dominant targets, large actions and deliberate spacing.	Focuses attention on the current learning step.
Flexibility & Efficiency	Mode selection and shared gameplay	Supports either gaze dwell or hand-pointer/pinch for the same task.	Accommodates different interaction preferences; efficiency requires later evaluation.

4.5 Additional HCI reasoning and implementation boundary

Norman's Gulf of Execution is addressed by explicit instructions ("Look at an object to select it" and "Pinch to Choose"). The Gulf of Evaluation is addressed by feedback such as Calibration Complete, Camera Ready and Great Job. Hick's Law motivates two clear mode choices rather than many competing options. Fitts's Law supports sufficiently large targets in calibration and gesture-controlled selection. Three-step tutorials limit extraneous cognitive load. Gaze and gesture are proposed natural input methods, while planned spoken guidance would complement the visual channel.

The Figma file currently demonstrates screen appearance and possible hotspot navigation only. Camera access, live object detection and tracking, gaze estimation/calibration, hand landmarks and pinch classification, selection validation, tracking-loss detection, recorded results and spoken feedback require a future implementation and study. The proposal therefore makes no claims of tested accuracy, efficiency or educational gains.

5. Context-Aware / Interaction Design

Section author: Youssef Khaled (247637)

5.1 Interaction concept

The proposed application is a camera-based “Find the Object” learning game for children aged 5–10. It recognizes and tracks physical objects and lets children select their on-screen representations through one of two independent techniques: gaze dwell or a hand-controlled pointer with pinch confirmation. The shared task, object display, feedback, and rules remain consistent between the modes.

Primary gameplay does not require mouse, keyboard, or touchscreen input. Adult assistance is permitted for initial permissions and camera placement; hands-free setup/navigation paths are intended for later implementation.

5.2 Architecture and information flow

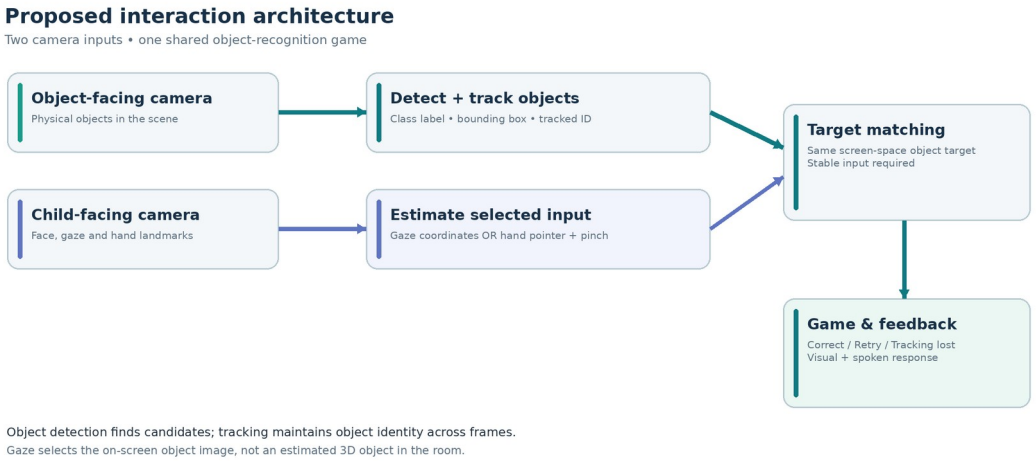


Figure 6. Proposed two-camera sensing architecture (Youssef).

Component	Role in proposed system
Object-facing camera	Captures the physical objects and their arrangement.
Detection & tracking	Proposes object classes and bounding boxes; maintains target identity across frames.
Child-facing camera	Supports facial/gaze estimation in Eye Tracking mode or hand landmark detection in Gesture mode.
Target matching	Maps the selected gaze point or gesture pointer to an eligible tracked object in the displayed camera view.
Selection & feedback	Validates dwell or pinch, checks the requested object class, and presents a clear visual/audio response.

5.3 Eye Tracking interaction

The child-facing camera estimates where the child is looking on the screen. The system does not claim to identify directly which physical object the child looks at in the three-dimensional room; gaze selection refers to the object's image in the live camera display.

1. Show simple calibration dots and estimate the mapping from observed eye/face features to display coordinates.

2. Display tracked objects and give a spoken task prompt, for example “Can you find the apple?”

3. Match the estimated gaze point to the on-screen bounding/selection region of an eligible tracked target.

4. Begin a visible dwell-progress indicator while the child looks at the same stable target. The starting proposal is 1.5 seconds, subject to pilot testing.

5. Reset or pause progress if gaze confidence drops, the child looks away, or target tracking becomes ambiguous; do not accept an unreliable selection.

6. When dwell completes, confirm the selected object and display correct-answer or gentle-retry feedback.

5.4 Hand Gesture interaction

1. Display a short tutorial showing how to position the hand, move the pointer, and pinch.

2. Detect hand landmarks in the child-facing camera and map hand motion to a visible on-screen pointer.

3. Highlight one candidate target when the pointer enters its tracked region.

4. Interpret a deliberate thumb-index pinch as the selection event only when one valid object is targeted.

5. Ignore duplicate selection events until the hand releases and pinches again; temporarily disable selection when tracking is uncertain.

6. Compare the selected tracked object against the requested category and provide immediate feedback.

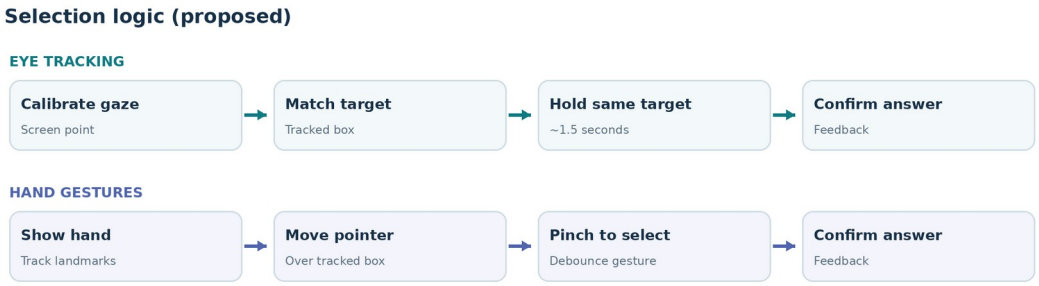


Figure 7. Shared validation with input-specific selection (Youssef).

Both interaction techniques operate on the same live object scene. Object tracking preserves the selected target ID across consecutive frames so transient detection changes are less likely to change the intended answer.

5.5 Context awareness and recovery

Context awareness means the proposed application notices relevant sensing and environmental conditions, then adapts its guidance or disables unreliable input. Lecture 3 distinguishes physical, social and organizational context in PACT. Here, physical factors include lighting and camera placement, while social and organizational factors include adult assistance, classroom supervision and responsible handling of children's camera data. The responses below are design proposals, not functions already implemented in Figma.

Condition	Potential signal	Planned response
Low light or glare	Poor image quality or repeated detection failure	Ask for brighter, more even lighting and show an example of suitable positioning.
Camera permission denied / unavailable	No camera stream	Pause setup and ask a supervising adult to check the permission or connection.
Objects not recognized	No eligible detection boxes	Ask the child/adult to bring a supported object into view; do not start a question without a target.
Occluded or overlapping objects	Ambiguous bounding boxes / tracked IDs	Ask to separate the objects; suppress selection where target identity is uncertain.
Object moved or temporarily lost	Track confidence drops or tracked target disappears	Suspend selection progress; reacquire a stable track before resuming.
Face or eyes not visible	Unavailable gaze estimates	Pause eye mode; show an age-appropriate prompt to face the camera.
Gaze calibration unsuccessful	Low calibration quality / unstable gaze mapping	Offer a calm retry or an adult-assisted switch to the gesture mode.
Hand outside camera view	Missing landmarks / hand confidence	Display "Show your hand to the camera" and disable pinch selection.

Condition	Potential signal	Planned response
Accidental or unstable input	Short gaze hits or repeated pinch transitions	Require stable dwell/valid pinch and release before the next command.

5.6 Interaction safeguards

Target eligibility: only tracked objects that correspond to supported labels become selectable; ambiguous overlaps should not trigger a guess.

Input gating: when sensing is lost, stop dwell progress or gesture acceptance rather than recording an incorrect answer.

Child-friendly recovery: provide short, supportive audio and visual instructions (“Let's try again”) without penalties for camera or tracking errors.

Navigation: design an equivalent gaze/gesture way to pause, retry and proceed; any adult-only setup controls should be identified separately.

5.7 HCI justification

The interaction design draws on the seven usability rules presented in HCI Lab 2. Ahmed's application-design section documents the visual implementation; the points below explain the interaction-level rationale.

Lab 2 principle	Interaction design decision	Why it matters
Navigation	Shared progression from tutorial/calibration to setup, gameplay, feedback and results.	The child can predict what happens next in either mode.
Familiarity	Familiar object names; a visible pointer, gaze indicator and simple task prompts.	Avoids requiring technical knowledge or gesture memorization.
Consistency	Same tasks, scene, answer checking and feedback between modes.	Reduces relearning and supports a fair comparison.
Error Prevention	Dwell threshold, pinch confirmation, tracking-quality gating, no duplicate pinch.	Reduces unintended selection and false confidence.
Feedback	Hand detected, camera ready, visible selection progress, correct/retry messages.	Closes the Gulf of Evaluation: the child knows whether the action worked.
Visual Clarity	One central task with readable instruction and limited secondary information.	Helps manage unnecessary cognitive load.
Flexibility & Efficiency	Two independent interaction modes; brief practice and optional retries.	Supports different abilities without promising equal performance.

Lecture 1’s Gulf of Execution motivates clear tutorial and calibration cues; the Gulf of Evaluation motivates visible tracking status and confirmation feedback. Lecture 2’s Cognitive Load Theory supports brief spoken prompts and limited on-screen distractions. These ideas justify design choices, not proven improvements in child performance.

Lecture 4 adds motor-control considerations: Fitts’s Law motivates generous, well-separated pointer targets, Hick’s Law supports presenting few choices at once, and Steering Law discourages narrow movement paths. These are design heuristics here, not numerical predictions for webcam gaze or children’s hand movement. The lecture’s speed–accuracy discussion also warns against judging the faster technique as better without examining errors and recovery.

5.8 Proposed comparison methodology

In a later supervised study, children with guardian consent and age-appropriate assent could complete equivalent object-finding tasks in both modes. Counterbalance mode order to reduce learning/fatigue effects; keep prompts, displayed objects, number of choices, feedback and target layout comparable. Record calibration

and tutorial time separately from question-response time, and log detection/tracking interruptions so input failures are not mistaken for children's mistakes.

Measure	Proposed operational definition
First-attempt correctness	Percentage of questions with the requested object selected on the first valid attempt, by technique.
Completion time	Time from each spoken/visual prompt to a correct selection, including retries; separately report time to first attempt.
Setup burden	Tutorial/calibration time and number of failed setup or recalibration attempts, reported separately by mode.
Errors and reliability	Wrong-object choices, unintended selections, tracking loss, retry count and recovery time; distinguish system errors from user errors.
Usability / satisfaction	Brief age-appropriate ease/enjoyment responses, with parent/teacher observations where suitable.
Engagement	Observed willingness to continue and session completion; avoid inferring attention or feelings from a camera alone.

5.9 Limitations, ethics and planned validation

Phase I establishes a design and evaluation plan only. No model accuracy, child usability advantage, or tracking reliability has yet been demonstrated.

Webcam gaze estimation: accuracy depends on calibration, head movement, lighting and screen geometry; the 1.5-second dwell target remains a starting assumption.

Hand interaction: fingertip-to-screen mapping, pinch thresholds, gesture fatigue and accidental selection need prototype testing.

Camera hardware: the proposed child-facing and object-facing setup is not yet verified. Synchronization and field-of-view constraints may require changes.

Detection and tracking: object categories, box quality, temporary occlusion and track continuity must be evaluated before claiming support for a given object.

Fairness: gaze dwell and pinch differ in setup effort, motor demands and error modes. Report first-attempt correctness, recovery and calibration burden alongside speed, and consider age-related differences without assuming one technique suits every child.

Child safeguards: conduct any child study with appropriate university approval, guardian permission and suitable supervision. The design should avoid collecting unnecessary identifiable footage or storing gaze/face information by default.

Accessibility: use spoken guidance, high contrast, sufficiently large target regions and age-appropriate pacing; test these assumptions with representative users.

5.10 Alignment with the Phase I Figma prototype

Ahmed Wahba's Figma design depicts: Welcome; Interaction Mode Selection; Hand Gesture Tutorial; Eye Tracking Calibration; Camera & Object Setup; Main Gameplay; Feedback States; and Session Results. The planned environmental responses map chiefly to camera readiness, calibration, gameplay tracking and Tracking Lost / Camera Unavailable feedback states. These Figma screens illustrate the same proposed interaction logic.

Technical confidence scores shown in the illustrative prototype are for design demonstration; they should not be presented as child-facing performance measurements. The reported design remains a simulated Figma flow rather than a working camera application.

• MSA University, CS484 HCI Project (Phase I), Fall 2026: grading criteria and team form.

• MSA e-Learning, Phase I assignment description (screenshot supplied by course team): required deliverables and one-PDF submission.

• HCI Lab 2 (Fall 2026), slides 19–25: Navigation, Familiarity, Consistency, Error Prevention, Feedback, Visual Clarity, Flexibility & Efficiency.

• HCI Lecture 1 (Fall 2026), slides 13–15: interaction laws and Norman's Gulfs.

• HCI Lecture 2, The Multi-Modal Transition, slides 2–6: natural interfaces, gaze/gesture, cognitive load and visual attention.

• HCI Lecture 3, User Research Framework & PACT Analysis (Fall 2026), pp. 14–24: PACT contexts, technologies and links to interface requirements.

• HCI Lecture 4, Physical Laws in HCI (Fall 2026), pp. 4–18, 19–30 and 41–52: Fitts's Law, target accuracy, Steering Law, Hick's Law and motor-control considerations.

• Ahmed Wahba, Application Design & HCI Principles (Phase I working report and Figma design, October 2026).

References and project links

Course materials

MSA University. CS484 Human-Computer Interaction, Fall 2026. Lecture 1: HCI foundations and user-centered design.

MSA University. CS484 Human-Computer Interaction, Fall 2026. Lecture 2: Multimodal interaction and cognitive considerations.

MSA University. CS484 Human-Computer Interaction, Fall 2026. Lecture 3: User Research Framework and PACT Analysis, especially pp. 14–24.

MSA University. CS484 Human-Computer Interaction, Fall 2026. Lecture 4: Physical Laws in HCI.

MSA University. CS484 Human-Computer Interaction, Fall 2026. Lab 2: HCI usability principles.

MSA University. CS484 Human-Computer Interaction, Fall 2026. Lab 3: Survey design guidance, especially slides 15–20.

MSA University. CS484 HCI. Phase I project instructions (application design, principles, PACT, scenario, and survey/critique).

Team-produced source documents

Mostafa Mohamed (249141). Scenario & User Journey, revised edition, October 2026.

Malak Gamal (247419). PACT Analysis, October 2026.

Mohamed Ramadan (246813). Survey & Critical Analysis, October 2026.

Ahmed Wahba (245559). Application Design & HCI Principles, reviewed/edited version, October 2026.

Youssef Khaled (247637). Context-Aware / Interaction Design, version 3, October 2026.

Design and coordination links

Figma — Interactive Learning Assistant

https://www.figma.com/design/ldE4Mbl0MEigqCEDxKZZfo/Interactive-Learning-Assistant-for-Children?node-id=0-1

GitHub — Interactive Learning Assistant HCI

https://github.com/youssefkhaled222/interactive-learning-assistant-hci

Trello — HCI Project Fall 2026

https://trello.com/b/lo6uzFCW/hci-project-fall-2026

Note: GitHub and Trello are private team workspaces. Access may need to be granted separately if the instructor wishes to inspect them.

Appendix A — Survey form settings and branching

The full questionnaire, including all thirteen questions and the rationale for each, appears in Section 3. This appendix preserves the key information needed to reproduce the proposed survey if the team later publishes it in Google Forms.

Form information and privacy

Title: Find the Object: Parent and Teacher Survey

Respondents: Parents/guardians of children aged 5–10 and teachers of that age range. Participation is voluntary. The form does not ask for names, email addresses, or identifying information about children. Respondents may stop at any time. The open comment at Q13 is optional.

Recommended settings

- Disable automatic email collection; do not require participants to sign in solely to limit responses.
- Show a progress bar; make Q1–Q12 required, Q13 optional, and require one answer for each Q4 grid row.
- Limit Q5 and Q11 to a maximum of three selections, with an “Other (please specify)” option.
- For Q6, “None of these” and “Not sure” are exclusive and should never be combined with specific available devices.
- Q11 asks which accessibility needs should be investigated first, not which children deserve support.
- Treat Q8 and Q9 responses as expectations, not results of a comparison study.

Section navigation and conditional branching

Form section	Questions	Branching
1. About you	Q1	“None of the above” ends form; eligible respondents continue.
2. About the child	Q2–Q3	“0 days” educational-app use skips Q4 and moves to section 4.
3. Current app use	Q4	Otherwise completed before section 4.
4. Setting and preferences	Q5–Q7	Continue to technique comparison.
5. Two selection methods	Q8–Q9	Provide equally worded descriptions of both input techniques.
6. Accessibility and privacy	Q10–Q13	Submit form after final optional comment.

The survey had not been published or distributed at the time the section was prepared; no responses are reported. No unprovided Google Apps Script is required or referenced in this integrated document.